

Lesson 25 — Design Challenge — Scorekeeper Game

Matches LEGO Teacher Portal lesson: "Dinner Decider".

Objective (I can...): I can design an interactive game or gadget that stores and uses data. **CS concept:** Applying variables with events, loops, and conditionals · **Builds on:** Lessons 21–24. *Also serves as flex/catch-up if the unit ran long.*

AI Connection (2 min): A game AI tracks your score and adapts to it — data drives its choices. *Impacts prompt:* "A game that adapts to you can be fun — who does that help, and could it ever bother a player?"

Materials: LEGO CS & AI kit + device per pair, plan sheets, trigger objects. **Model already built** (Variables model from Lesson 21) — the 30 min is plan + code + test, small reconfigs only.

60-minute flow - Warm-up (8): Variables review — What is a variable? How do you change it? How can it make a decision? - **Concept + demo (10):** Challenge: build a game/gadget that keeps score or counts and reacts when it hits a target. Coach models a plan with one variable, then hands off. - **Build/Code (30):** Pairs plan and code a program using at least one variable plus an event and a conditional (a win/lose or milestone reaction) on the built model. *Swap at 15 min.* - **Challenge (7):** Add a second variable, a reset, or a timer. - **Share + pack-up (5): Toy-fair game day** — pairs rotate to actually play each other's games, then each creator explains the variable behind the scoreboard. **Photograph each build.**

Pair roles: Co-plan; Driver codes / Navigator checks the variable logic; swap at 15 min. *(Trio rotates plan/code/test.)*

Differentiation: *Support* — a counter-only target with a partial plan. *Extension / fast-finisher* — require two variables and a win condition. *Reteach* — reuse a Lesson 23–24 program as a template.

Exit check: Pair delivers a working game/gadget that stores, updates, and uses a variable in a decision.

Progress layer: Send home the **Unit 5 parent note** ("your child can build a game that keeps score and reacts") with the **build photo**. **At-home (optional):** Sketch your game's "scoreboard"; label what it tracks. ~10 min.
